

Time: 180min

Closed Book

Max Marks: 80

Average: 13.5 / 80

- i. Start every question on a new page.
- ii. Please enclose all final answers in a box. All answers must be written in pen only. Answers in pencil will not be evaluated.

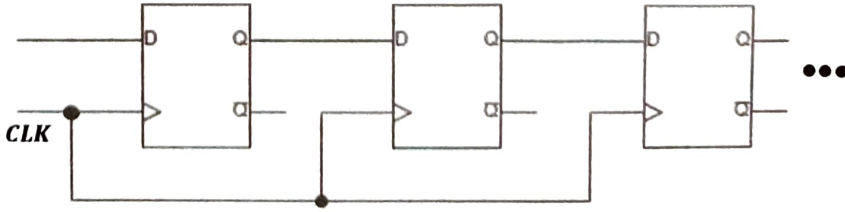
1. (10 marks) Implement the function $F = \overline{CD + A(B + E)}$ with a complex static CMOS gate with matched worst case rise and fall resistance which is half that of the smallest inverter in the technology. Assume that the smallest inverter in this technology is sized with $(W/L)_p = 3$ and $(W/L)_n = 1$. Your design must minimize the parasitic cap at the output node.
 - a. Draw the schematic of the gate and size the transistors
 - b. Assuming that there is no reduction in capacitance due to sharing or contact, find the delay of the following transition and express in multiples of τ_{inv} .
 - i. (ABCDE): (10000) $\rightarrow$ (10001)
 - ii. (ABCDE): (11011) $\rightarrow$ (01011)
2. (8 marks) A ring oscillator consisting of 15 minimum-sized (1.5:1) inverters placed on a 45nm chip, has a measured frequency of 3.17 GHz
 - a. What is the delay of a single stage of the oscillator?
 - b. If the RO performance is assumed to be representative of the transistor performance on the chip, what is the FO4 delay of the inverters on this chip?
 - c. If it is known that the rising delay of the inverter is 15% longer than the falling delay, what is the duty cycle of the signal from the ring oscillator?
3. (12 marks) In order to implement OR4 logic to drive a 6.8 fF on-chip capacitance, we end up with two options, (A) NOR4(1X) + INV(5X) or (B) NOR2(2X) + NAND2(? X). Assume the reference is a 2/1 sized inverter, and the gate and diffusion capacitance per unit width is $C = 1.25 \text{ fF}/\mu\text{m}$. Calculate the size ratio of NAND2 in option (B) to make the two options achieve the same delay. Assume the minimum CMOS transistor width $(W_p + W_n)$ is 80nm. The convention NOR2(BX) means that the transistors in the NOR2 gate are sized B times larger than the smallest NOR2 gate in the technology.
4. (9 marks) Consider an inverter sized for equal rise and fall delays. The technology is a $0.25\mu\text{m}$ technology with $V_{DD} = 2.5 \text{ V}$, $V_{Tn} = -V_{Tp} = 0.6 \text{ V}$, $k'_n = 3k'_p = 120 \mu\text{A}/\text{V}^2$. The inverter is connected to a large wire load of 150pF. Neglect the drain caps of the transistors.
 - a. Find the falling delay
 - b. Find the noise margins of the circuit
 - c. Redesign the circuit so that the falling delay is reduced to 50% of the value in (a) without any change in the rising delay
 - d. How will the noise margin of the circuit in (c) compare to the values you found in (b). Be specific in the highlighting the difference by plotting both VTC on the same axes.
5. (8 marks) Consider the register circuit in Fig.5. The flip-flops have the following characteristic timings: $t_{setup} = 100 \text{ ps}$, a maximum CLK-to-Q delay of 150ps and a minimum CLK-to-Q delay of 100ps.
 

Figure 5

- a. How fast can this circuit be clocked at? Does this depend on the size of the register?

- b. What is the limit on the hold time of the flop at this frequency? Does it depend on the size of the register?
- c. Suppose we now want to add functionality to the register through a MUX that enables load, reset, complement option at the inputs of every flop where the fastest path is 60ps and the slowest path is 130ps between the 2 flip flops, would your answers to part(a) and (b) change? If yes, give their new values.
- d. How would the answer in (c) change if a maximum CLK skew of 30ps can exist on the chip.
6. (11 marks) Consider a NAND2(2X) gate driving another (6X) NAND2 gate along a metal line of length l whose resistance and capacitance are given by $100 \Omega/\mu m$ and $0.18 fF/\mu m$ respectively. The delay of the smallest unloaded inverter in this technology is 25ps. The PMOS is sized 3 times the NMOS in the smallest inverter for matched resistance of $5 k\Omega$. The outer inputs of the 1st NAND gate is connected to V_{DD} .
- a. Plot the delay from the input of the 1st NAND gate to the input of the 2nd NAND gate as a function of l .
- b. What is the maximum allowable value of l , to ensure that this delay is below 4 FO4 delays in this technology?

7. (12 marks) Size the path in Fig. 7 for minimum possible delay and express the min delay in ps. It is known that the delay of the smallest inverter with a fanout of 4 in this technology is 70ps. The smallest inverter is sized $P/N=3/2$ for matched output resistance for output HI and LO states. The input cap is $6C$ and the output capacitance is $800C$ where C denotes the gate and the drain capacitance of the NMOS of size 1.

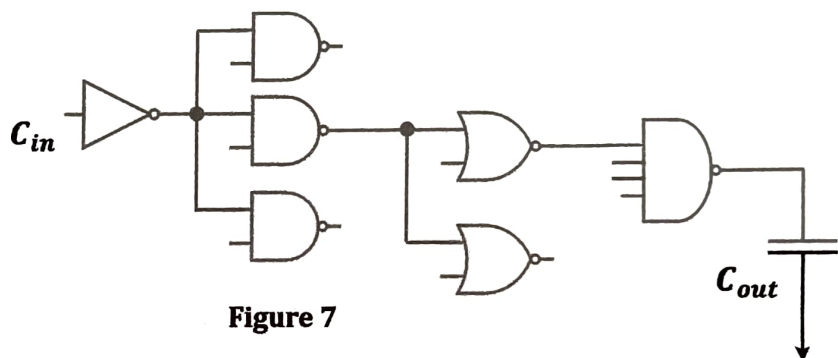
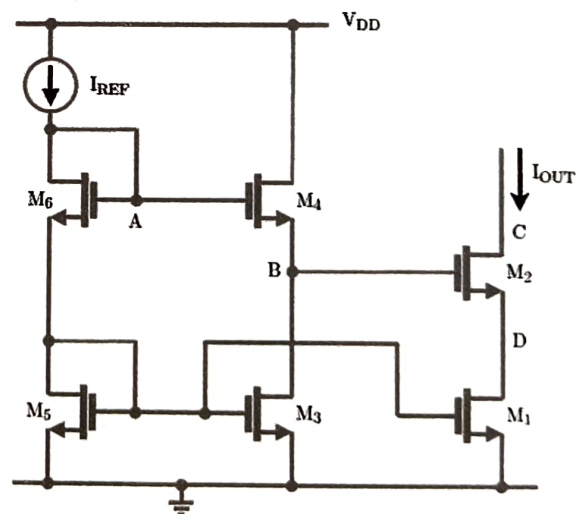


Figure 7

8. (10 marks) The regular cascode current mirror (not shown here) has a disadvantage in terms of large minimum voltage across output. The modified low voltage cascode current mirror is shown below where minimum voltage across output (at point C in figure) is less by V_{Tn} in comparison to regular cascode current mirror. Assume that all transistors are in saturation. If $I_{OUT} = I_{REF} = 50 \mu A$, minimum voltage across regular cascode current mirror is 0.7 V, $I_{D1} = I_{D3} = I_{D5}$ and $(W/L)_1 = (W/L)_2$. Given $V_{DD} = 1.8 V$, $k'_n = 100 \mu A/V^2$, $V_{Tn} = 0.45 V$, Find

- a. All transistor sizes
- b. Minimum voltage at points A, B and D.



*** All the best ***